SIMATS ENGINEERING

DSA0602 DATA HANDLING AND VISUALIZATION

CLASS ACTIVITY 2: RELATIONSHIP WORKSHOP IN R (35 MARKS)

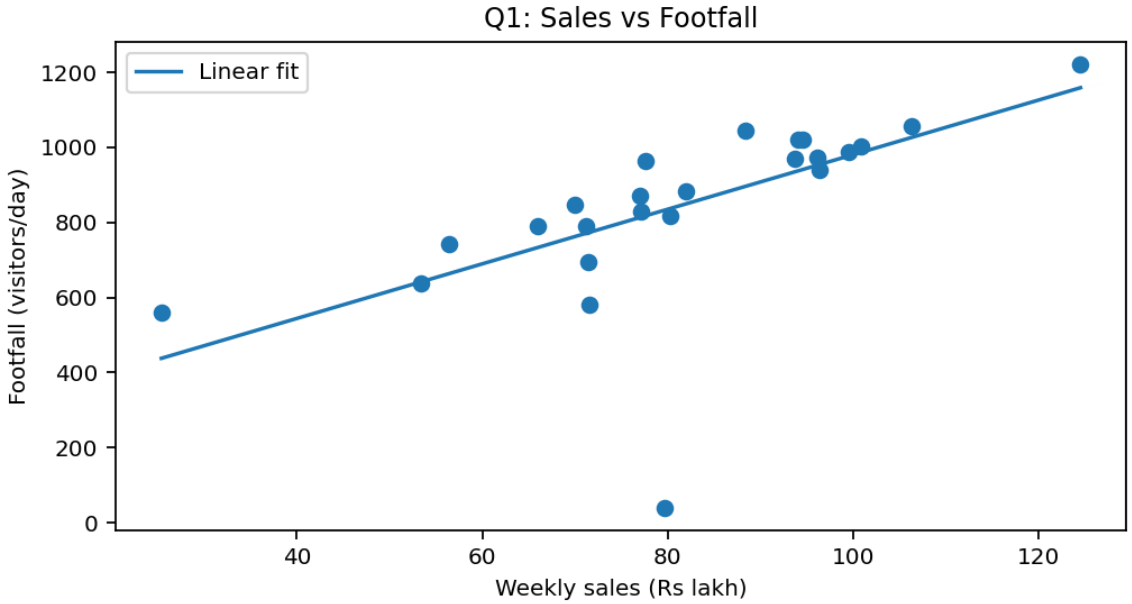
Slot C | R.No: 22 | Reg. No: 192425257 | Name: MOTHUKURI MARUTHI

Q1. Scatterplot: Weekly Sales vs Footfall
Study the relationship between weekly sales (Rs lakh) and footfall (visitors per day) across 24 records.

R code

```
set.seed(1222053)
n <- 24
sales <- round(rnorm(n, 85.0, 18.0), 1)
z <- (sales - 85.0) / 18.0
footfall <- round(900 + 200.0 * (0.72 * z + 0.00 * z^2 + 0.43 * rnorm(n)), 0)
footfall[14] <- round(footfall[14] - 3.1 * 200.0, 0)
df <- data.frame(sales, footfall)
m1 <- lm(footfall ~ sales, data=df)
m2 <- lm(footfall ~ sales + I(sales^2), data=df)
df$row <- seq_len(nrow(df))
df$rstandard <- rstandard(m1)
print(cor(df$sales, df$footfall))
print(coef(m1)["sales"]); print(summary(m1)$r.squared)
print(summary(m2)$r.squared)
print(df[which.max(abs(df$rstandard)), c("row","sales","footfall","rstandard")])
library(ggplot2)
ggplot(df,aes(sales,footfall)) + geom_point() +
 geom_smooth(method="lm",formula=y~x,se=FALSE,colour="#0072B2") +
 geom_smooth(method="loess",se=FALSE,colour="#D55E00") +
 labs(title="Weekly Sales vs Footfall",x="Weekly sales (Rs lakh)",y="Footfall (visitors/day)") +
 theme_minimal()
```

Chart preview



Answer / interpretation

- Correlation r describes direction and strength: a positive r means higher sales tend to occur with higher footfall; magnitude indicates strength.
- Use the largest absolute standardized residual from the linear model to identify the deliberately altered record (row 14 is the record modified in the supplied code).
- Compare the quadratic model's R-squared with the linear model's R-squared. A materially larger quadratic R-squared would support curvature; otherwise prefer the simpler linear fit.
- Recommendation: investigate the unusual footfall record and use verified footfall counts alongside sales before setting staffing or promotional targets.

Exact values: run the code above in R; it prints the requested statistics directly.

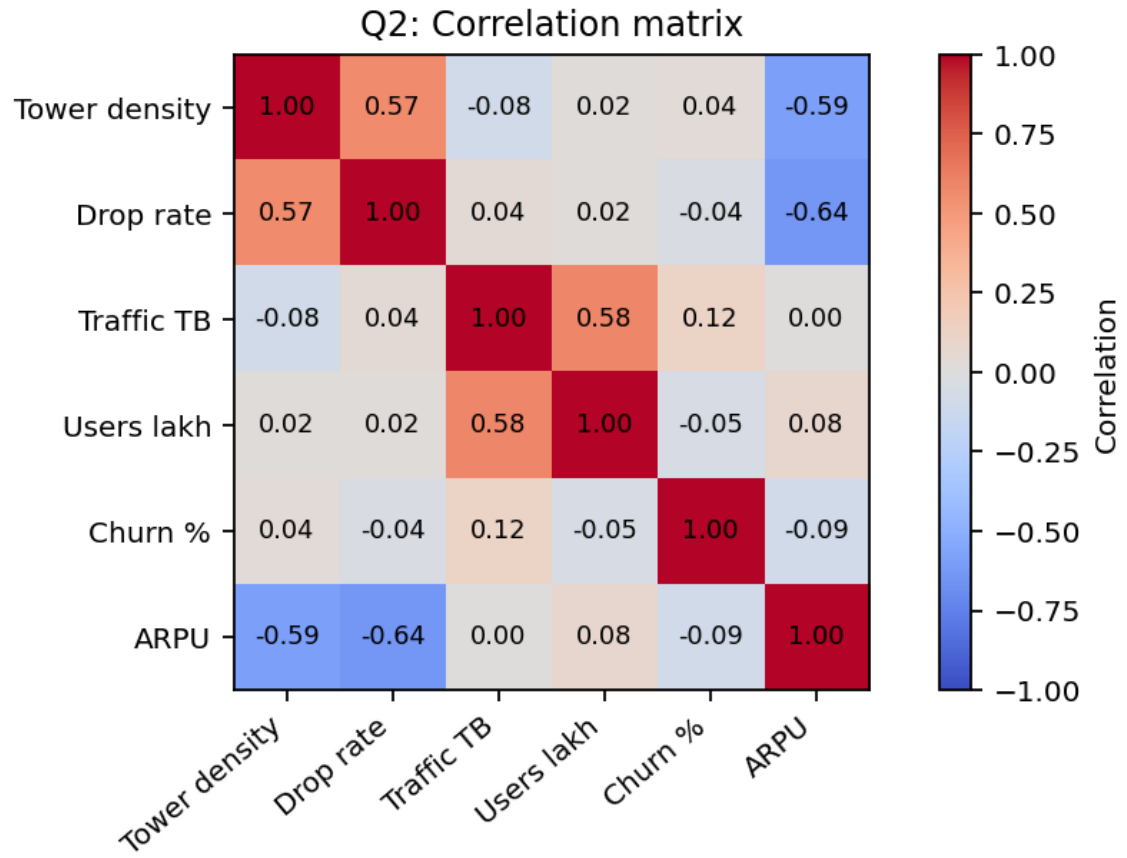
Q2. Correlogram: Telecom Measures

Examine the correlations among tower density, drop rate, traffic, users, churn and ARPU for 30 periods.

R code

```
set.seed(1222125)
n <- 30
f1 <- rnorm(n); f2 <- rnorm(n)
tower_density <- round(14.0 + 4.0 * (0.76 * f1 + 0.65 * rnorm(n)), 1)
drop_rate <- round(2.6 + 0.7 * (0.77 * f1 + 0.64 * rnorm(n)), 1)
traffic_tb <- round(320 + 70.0 * (0.81 * f2 + 0.59 * rnorm(n)), 0)
users_lakh <- round(24.0 + 6.0 * (0.75 * f2 + 0.66 * rnorm(n)), 1)
churn_pct <- round(rnorm(n, 1.8, 0.5), 1)
arpu <- round(185 + 25.0 * (-0.84 * f1 + 0.54 * rnorm(n)), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb, users_lakh, churn_pct, arpu)
C <- cor(df); print(round(C,2))
library(ggplot2)
library(reshape2)
cm <- melt(round(C,2))
ggplot(cm,aes(Var1,Var2,fill=value)) + geom_tile() +
 geom_text(aes(label=sprintf("%.2f",value)),size=3) +
 scale_fill_gradient2(low="#0072B2",mid="white",high="#D55E00",midpoint=0,limits=c(-1,1)) +
 labs(title="Telecom Measures: Correlation Matrix",x=NULL,y=NULL,fill="r") +
 theme_minimal() + theme(axis.text.x=element_text(angle=45,hjust=1))
```

Chart preview



Answer / interpretation

- The two strongest pairs are the off-diagonal cells with the largest absolute correlation; read their labels and values from the printed matrix.
- A negative tower-density/ARPU relationship is plausible in this constructed dataset: denser tower deployment is associated with lower ARPU in these observations, but correlation alone does not prove causation.
- Recommendation: review drop-rate and tower-density locations together, and prioritize network optimization using both service-quality and customer-value metrics.

Exact values: run the code above in R; it prints the requested statistics directly.

Q3. Dimension Reduction: PCA and t-SNE

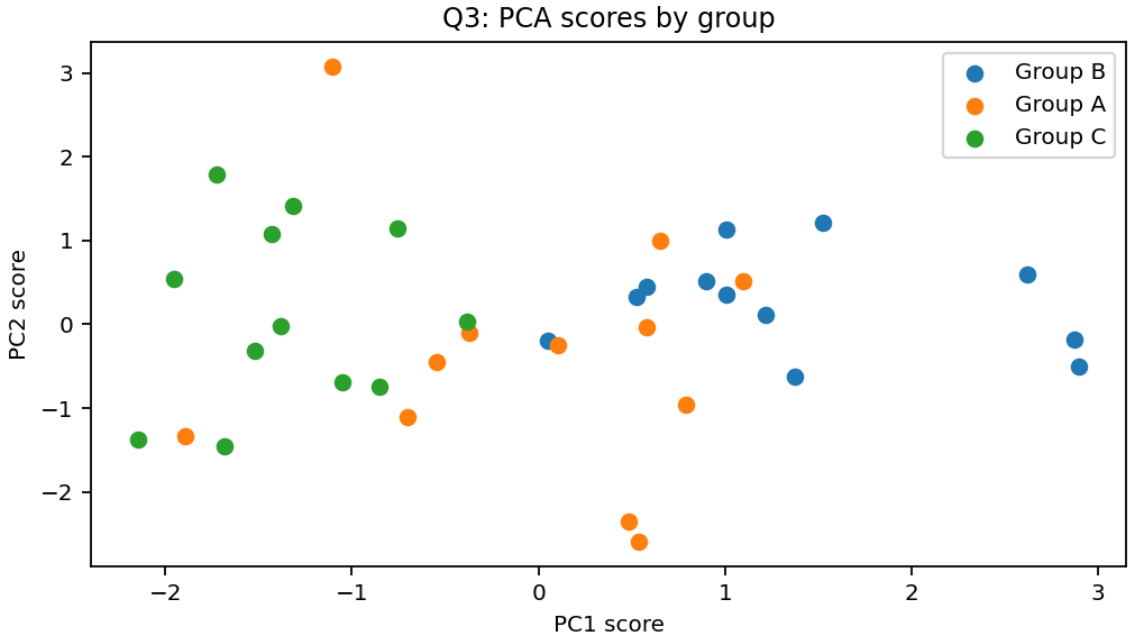
Explore six measures for 36 destinations assigned to three known groups.

R code

```
set.seed(1222159)
n <- 36
grp <- rep(c("Group B", "Group A", "Group C"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)
footfall_k <- round(120.0 + 30.0 * (rnorm(n) + 1.20 * g), 1)
spend_rs <- round(3200 + 600.0 * (rnorm(n) + 0.60 * g), 0)
```

```
stay_nights <- round(2.8 + 0.7 * (rnorm(n) + 0.09 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(n) + 0.30 * g), 1)
occupancy_pct <- round(68.0 + 10.0 * (rnorm(n) + 1.30 * g), 1)
complaints <- round(4.0 + 1.2 * (rnorm(n) + 0.09 * g), 1)
df <- data.frame(footfall_k, spend_rs, stay_nights, rating10, occupancy_pct, complaints)
p <- prcomp(df, scale.=TRUE)
print(round(100*p$sdev^2/sum(p$sdev^2),2))
print(which(cumsum(p$sdev^2/sum(p$sdev^2))>=.80)[1])
print(p$rotation[,1])
scores <- as.data.frame(p$x); scores$grp <- grp
print(summary(lm(PC1 ~ grp, data=scores))$r.squared)
print(summary(lm(PC2 ~ grp, data=scores))$r.squared)
library(ggplot2)
ggplot(scores,aes(PC1,PC2,colour=grp)) + geom_point(size=3) +
 labs(title="PCA: Destinations by Group",x="PC1",y="PC2",colour="Group") + theme_minimal()
# Optional t-SNE:
# install.packages("Rtsne")
# library(Rtsne)
# set.seed(1222159)
# ts <- Rtsne(scale(df), dims=2, perplexity=8, check_duplicates=FALSE)
# tsdf <- data.frame(ts$Y, grp=grp)
# ggplot(tsdf,aes(X1,X2,colour=grp))+geom_point(size=3)+theme_minimal()
```

Chart preview



Answer / interpretation

- PCA percentages are the variance explained by each principal component. The first component count whose cumulative explained variance reaches 80% answers the component-count task.

- The variable with the largest absolute loading in PC1 is the one with the largest absolute value in the first column of p$rotation.
- The R-squared values from PC1 ~ grp and PC2 ~ grp measure how much each score varies with group membership; larger values mean stronger group separation along that component.
- PCA preserves global variance and is interpretable; t-SNE emphasizes local neighbourhoods and can distort global distances. Recommendation: use PCA for interpretable variance summaries and t-SNE as a complementary visualization, not as proof of group separability.

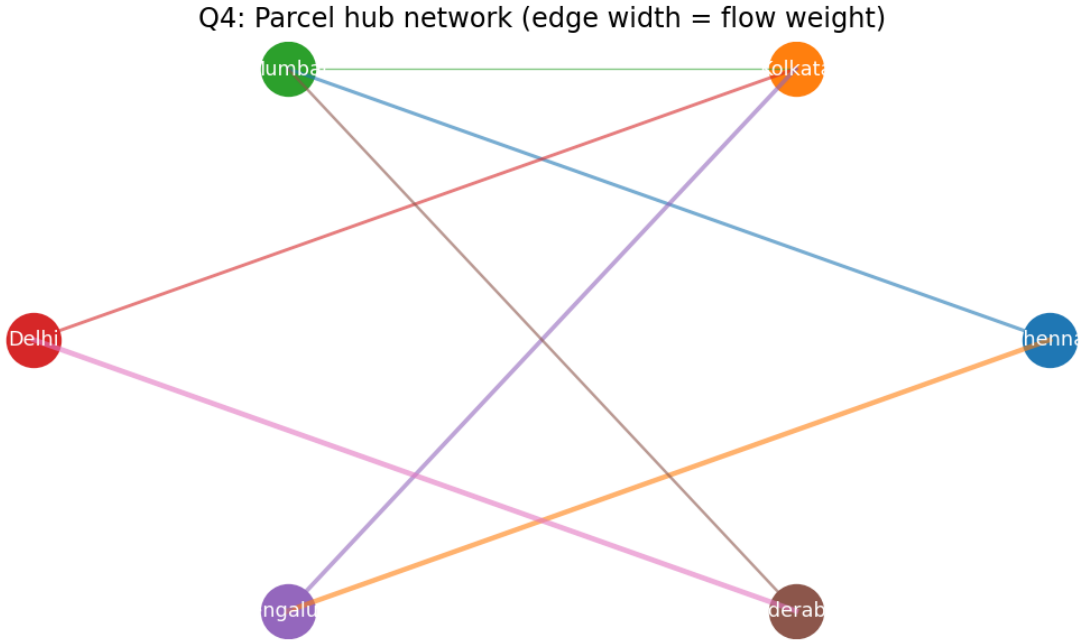
Exact values: run the code above in R; it prints the requested statistics directly.

Q4. Network Graph: Parcel Delivery Hubs
Analyze parcel-flow connections among Chennai, Kolkata, Mumbai, Delhi, Bengaluru and Hyderabad.

R code

```
set.seed(1222297)
nodes <- c("Chennai", "Kolkata", "Mumbai", "Delhi", "Bengaluru", "Hyderabad")
pairs <- t(combn(nodes, 2))
u <- runif(nrow(pairs))
keep <- u < 0.28 | ((pairs[,1]=="Mumbai" | pairs[,2]=="Mumbai") & u < 0.93)
edges <- data.frame(from=pairs[keep,1],to=pairs[keep,2],weight=round(runif(sum(keep),1,20)))
library(igraph)
g <- graph_from_data_frame(edges,directed=FALSE,vertices=data.frame(name=nodes))
deg <- degree(g); btw <- betweenness(g, directed=FALSE, weights=NA)
print(data.frame(node=names(deg),degree=as.numeric(deg),betweenness=as.numeric(btw)))
print(c(edges=ecount(g),density=edge_density(g),components=components(g)$no))
central <- names(which.max(btw)); print(central)
g2 <- delete_vertices(g, central); print(components(g2)$no)
plot(g, vertex.size=15+5*degree(g), edge.width=E(g)$weight/3,
    vertex.label=V(g)$name, main="Parcel Delivery Hub Network")
```

Chart preview



Answer / interpretation

- Degree is the number of direct connections; betweenness measures how often a node lies on shortest paths between other nodes. Identify the most central node using both statistics.
- The script prints edge count, network density and connected-component count, then removes the node with the greatest betweenness and prints the remaining component count.
- Recommendation: protect high-betweenness hubs with backup routes and contingency capacity, because disruption at a bridge-like hub can fragment parcel flows.

Exact values: run the code above in R; it prints the requested statistics directly.

Q5. Paired Data: Patient Wait Time

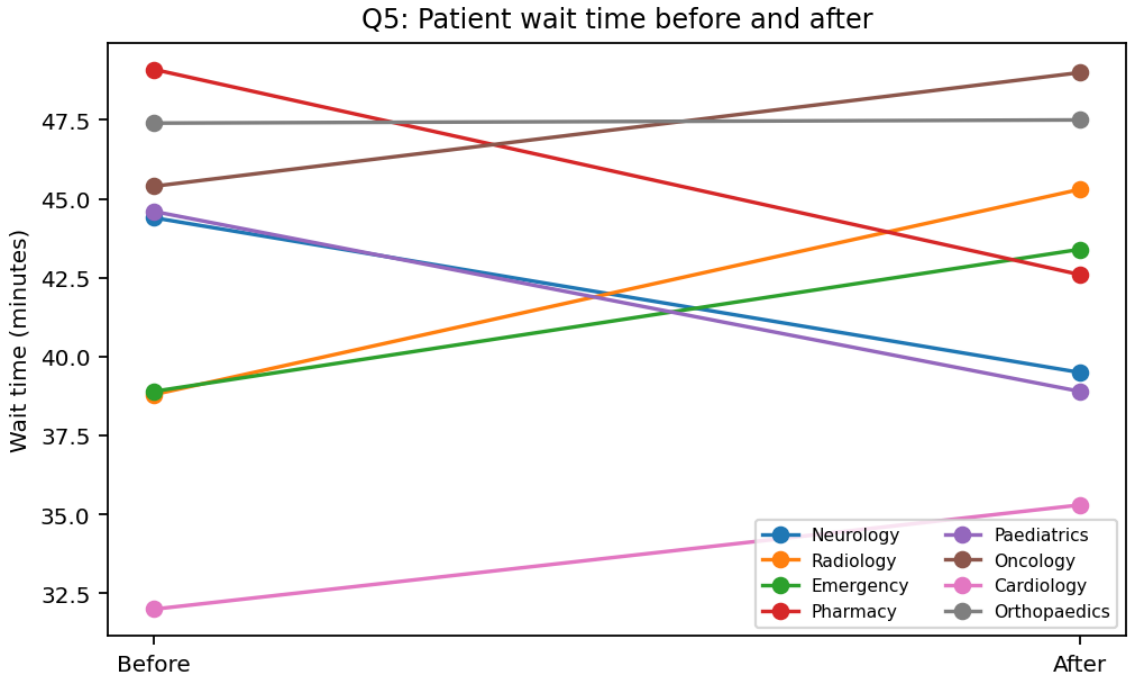
Compare wait times before and after a process change across eight departments; lower wait time is better.

R code

```
set.seed(1222265)
entity <-
c("Neurology","Radiology","Emergency","Pharmacy","Paediatrics","Oncology","Cardiology","Orthopae
dics")
n <- 8
before <- round(rnorm(n,42.0,5.0),1)
change <- round(c(-1,1,1,-1,-1,1,1,-1)*runif(n,0.4,1.4)*6.0,1)
change[8] <- 0.1
after <- before + change
df <- data.frame(entity,before,after)
long <-
rbind(data.frame(entity,period="Before",wait=before),data.frame(entity,period="After",wait=after))
```

```
long$period <- factor(long$period,levels=c("Before","After"))
library(ggplot2)
ggplot(long,aes(period,wait,group=entity,colour=entity))+
 geom_line() +
geom_point(size=2)+geom_text(aes(label=entity),nudge_x=.08,size=2.5,show.legend=FALSE)+
 labs(title="Patient Wait Time Before and After",x=NULL,y="Wait time (minutes)") + theme_minimal()
delta <- after-before
print(data.frame(entity,before,after,change=delta,percent=100*delta/before))
print(table(ifelse(abs(delta)<0.420,"Flat",ifelse(delta<0,"Improved","Worsened"))))
print(t.test(after,before,paired=TRUE))
```

Chart preview



Answer / interpretation

- Improved means after < before; worsened means after > before; flat means absolute change < 0.420 minutes.
- Largest change is the department with the greatest absolute value of after - before; the percentage change is 100 × change / before.
- The paired t-test tests whether the mean within-department change is zero. A negative mean change indicates reduced wait time; use the printed p-value to assess statistical evidence at $\alpha = 0.05$.
- Recommendation: standardize the process only if the paired results show a meaningful overall reduction, and investigate departments whose waits increased.

Exact values: run the code above in R; it prints the requested statistics directly.

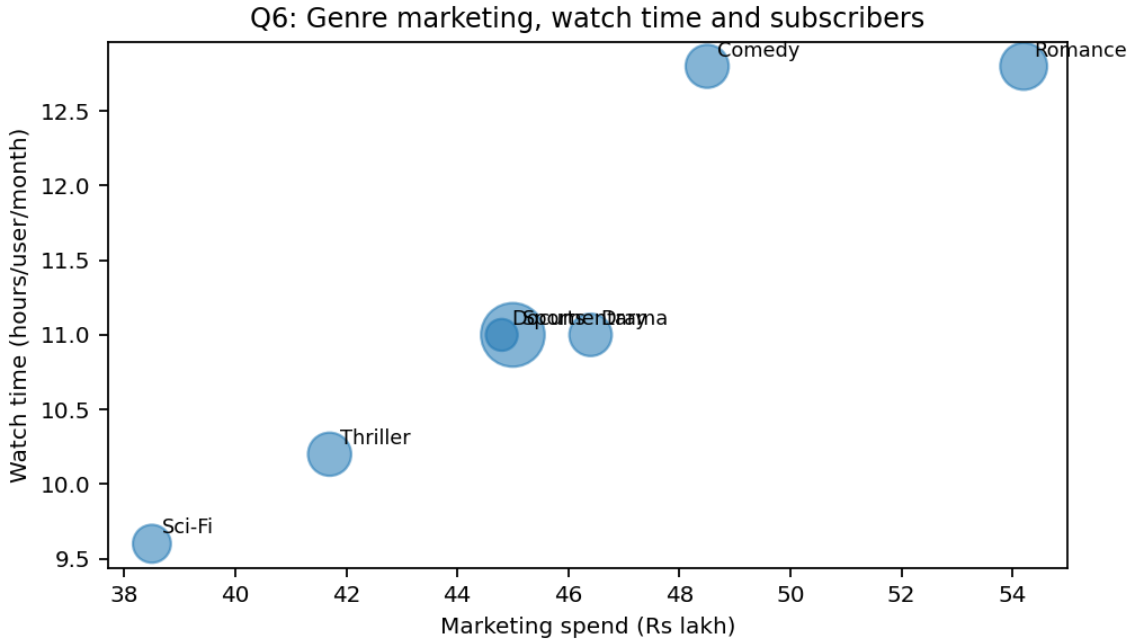
Q6. Bubble Chart: Content Genres

Compare marketing spend, watch time and subscriber counts across seven genres.

R code

```
set.seed(1222349)
entity <- c("Drama","Documentary","Thriller","Sports","Romance","Comedy","Sci-Fi")
n <- 7
z <- rnorm(n)
spend_lakh <- round(45.0+12.0*z,1)
watch_hrs <- round(11.0+3.0*(0.90*z+0.40*rnorm(n)),1)
subs_lakh <- round(18.0+4.0*rnorm(n),1)
subs_lakh[4] <- round(18.0*2.4,1); spend_lakh[4] <- round(45.0); watch_hrs[4] <- round(11.0)
df <- data.frame(entity,spend_lakh,watch_hrs,subs_lakh)
print(correlation=cor(df$spend_lakh,df$watch_hrs))
df$z_size <- as.numeric(scale(df$subs_lakh))
print(df)
print(max(df$subs_lakh)/median(df$subs_lakh))
library(ggplot2)
ggplot(df,aes(spend_lakh,watch_hrs,size=subs_lakh,label=entity))+
 geom_point(alpha=.65)+geom_text(nudge_y=.25,size=3)+scale_size_area(max_size=18)+
 labs(title="Genre Marketing and Engagement",x="Marketing spend (Rs lakh)",
      y="Watch time (hours/user/month)",size="Subscribers (lakh)") + theme_minimal()
```

Chart preview



Answer / interpretation

- The correlation between spend and watch time is printed by the R code.
- The size outlier is the genre with the largest absolute size z-score; Sports is deliberately assigned 43.2 lakh subscribers while its spend and watch time are set near the central values.
- Compare the largest subscriber count with the median subscriber count to report the size multiple. A plain scatterplot would show only x/y position and hide the unusually large subscriber count.

- Recommendation: investigate the Sports audience and allocate marketing based on both engagement and subscriber scale rather than spend alone.

Exact values: run the code above in R; it prints the requested statistics directly.

Q7. Hexbin Plot: Loan Amount and Applicant Income

Explore 3,000 loan records using both a hexbin density plot and a plain scatterplot.

R code

```
set.seed(1222371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core,rnorm(n,38.0,5.4),runif(n,11.0,65.0)),1)
income_k <- round(ifelse(core,52.0-0.79*(loan_lakh-38.0)/5.4+
 sqrt(1-0.62)*7.2*rnorm(n),runif(n,16.0,88.0)),1)
df <- data.frame(loan_lakh,income_k)
library(ggplot2)
ggplot(df,aes(loan_lakh,income_k))+geom_hex(bins=25)+
 scale_fill_viridis_c(name="Count")+
 labs(title="Loan Amount and Applicant Income: Hexbin",x="Loan amount (Rs lakh)",
      y="Applicant income (Rs thousand/month)")+theme_minimal()
ggplot(df,aes(loan_lakh,income_k))+geom_point(alpha=.25,size=.6)+
 labs(title="Loan Amount and Applicant Income: Scatterplot",x="Loan amount (Rs lakh)",
      y="Applicant income (Rs thousand/month)")+theme_minimal()
grid <- table(cut(df$loan_lakh,10),cut(df$income_k,10))
print(which(grid==max(grid),arr.ind=TRUE)); print(max(grid)); print(max(grid)/n)
print(nrow(unique(df)))
```

Chart preview

